

Weather Stream OGC API: Collections Examples

This document shows example requests for:

- adding a feature to the **sensors** collection
- retrieving data from a feature location

1. Add a feature

Use **POST** **/collections/sensors/items** with **Content-Type: application/geo+json**.

Example request

```
curl -X POST "http://localhost:5000/collections/sensors/items" \
-H "Content-Type: application/geo+json" \
-H "Accept: application/json" \
-d '{
  "type": "Feature",
  "geometry": {
    "type": "Point",
    "coordinates": [-75.123, 45.421]
  },
  "properties": {
    "sensor_id": "sensor-1001",
    "sensor_name": "Downtown Station",
    "available_variables": "temperature,humidity",
    "valid_start_datetime": "2026-05-01T00:00:00Z",
    "valid_end_datetime": "2026-05-31T23:59:59Z"
  }
}'
```

Expected result

- Status: **201 Created**
- Location header: URL of the created item, e.g.
http://localhost:5000/collections/sensors/items/1
- The feature is added to the **sensors** collection.

Verify the feature was added

The feature **id** as the item identifier, retrieve it directly:

```
curl "http://localhost:5000/collections/sensors/items/1?f=json"
```

You can also search the collection by a property value:

```
curl "http://localhost:5000/collections/sensors/items?
f=json&sensor_id=sensor-1001"
```

2. Retrieve data from the location of a feature

There are two related location-based endpoints in this API:

- **GET /collections/sensors/locations** lists available predefined locations
- **GET /collections/sensors/locations/{locId}** retrieves coverage data for one location

Step 1: list available locations

```
curl "http://localhost:5000/collections/sensors/locations?f=json"
```

Example result

The response should return one or more locations. Pick a location identifier from the result, for example **1**.

Step 2: retrieve coverage data for one location

```
curl "http://localhost:5000/collections/sensors/locations/1?f=json"
```

Retrieve a specific variable at that location

Request a single variable from the icechunkr repository, for example **2m_temperature**:

```
curl "http://localhost:5000/collections/sensors/locations/1?parameter-
name=2m_temperature&f=json"
```

Retrieve data for a time range or instant

```
curl "http://localhost:5000/collections/sensors/locations/1?datetime=2026-
05-12T12:00:00Z&f=json"
```

Or an interval:

```
curl "http://localhost:5000/collections/sensors/locations/1?datetime=2026-
05-12T00:00:00Z/2026-05-13T23:59:59Z&parameter-name=2m_temperature&f=json"
```

Expected result

- Status: **200 OK**
- Content type: **application/prs.coverage+json**
- Body: CoverageJSON payload for the requested location